

**KESELAMATAN DAN KESEHATAN KERJA (K3) BIDANG
PERTAMBANGAN BATU BARA ANALISIS MANAJEMEN RISIKO
DAN IDENTIFIKASI SUMBER BAHAYA PADA AREA KERJA**

PT. BUKIT ASAM TBK



Disusun Oleh:

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Jurusan Teknologi Informasi

D4 Sistem Informasi Bisnis

Politeknik Negeri Malang

2026

BAB I

PENDAHULUAN

1.1 Latar Belakang

Industri pertambangan batu bara merupakan salah satu sektor industri yang memiliki tingkat risiko kecelakaan kerja yang tinggi. Aktivitas seperti penggalian, pengangkutan, pengolahan, serta penggunaan alat berat dapat menimbulkan berbagai potensi bahaya bagi pekerja. Oleh karena itu, penerapan sistem keselamatan dan kesehatan kerja (K3) sangat penting untuk mencegah terjadinya kecelakaan kerja.

Menurut *International Labour Organization (ILO)*, setiap tahun jutaan pekerja di dunia mengalami kecelakaan kerja akibat kurangnya pengendalian bahaya di tempat kerja. Oleh karena itu, identifikasi sumber bahaya serta faktor yang mempengaruhi keselamatan kerja perlu dilakukan secara sistematis.

Pada sektor pertambangan batu bara, beberapa potensi bahaya yang sering ditemukan antara lain debu batu bara, kebisingan alat berat, penggunaan bahan kimia, serta kondisi kerja yang menimbulkan stres kerja. Penelitian menunjukkan bahwa paparan debu batu bara dapat menyebabkan gangguan sistem pernapasan pada pekerja tambang apabila terpapar dalam jangka waktu yang lama (Aini, 2015).

BAB II

PEMBAHASAN

2.1 Faktor-Faktor Pencegahan Kecelakaan Kerja

Pencegahan kecelakaan kerja merupakan hasil dari sistem keselamatan yang terintegrasi antara manajemen lingkungan kerja, dan pekerja.

1. Faktor Manajemen dan Organisasi

Faktor manajemen merupakan pondasi utama dalam pencegahan kecelakaan kerja.

Beberapa langkah pencegahan yang dilakukan antara lain:

- Komitmen manajemen terhadap program keselamatan kerja
- Penerapan Sistem Manajemen Keselamatan dan Kesehatan Kerja (SMK3)
- Penyusunan prosedur kerja atau SOP yang jelas
- Pelatihan keselamatan kerja secara berkala
- Investigasi kecelakaan kerja dan near miss

Contoh pada denah PT Bukit Asam:

- Ruang K3 / HSE Room (D) berfungsi sebagai pusat pengawasan keselamatan kerja.
- Pos pengecekan APD (A) memastikan pekerja menggunakan alat pelindung diri sebelum memasuki area kerja.

2. Faktor Teknis dan Lingkungan Kerja

Pencegahan kecelakaan kerja juga dapat dilakukan melalui pengendalian bahaya dengan metode hirarki pengendalian risiko, yaitu:

- 1) Eliminasi
- 2) Substitusi
- 3) Engineering control
- 4) Administrative control
- 5) Alat Pelindung Diri (APD)

Selain itu, pemeliharaan mesin dan peralatan kerja harus dilakukan secara berkala untuk memastikan kondisi mesin tetap aman.

Contoh pada denah PT Bukit Asam:

- Area produksi dan bengkel memerlukan pengawasan mesin dan peralatan kerja secara rutin.
- Jalur masuk dan keluar kendaraan tambang harus diatur dengan baik untuk mencegah kecelakaan kendaraan.

3. Faktor Individu atau Pekerja

Pekerja merupakan pihak yang langsung berinteraksi dengan sumber bahaya sehingga kepatuhan terhadap prosedur keselamatan kerja sangat penting.

Upaya pencegahan yang dapat dilakukan antara lain:

- Menggunakan alat pelindung diri (helm, sepatu safety, masker)
- Mematuhi prosedur kerja yang telah ditetapkan
- Mengikuti pelatihan keselamatan kerja
- Menjaga kondisi kesehatan dan kebugaran kerja

2.2 Faktor-Faktor yang Mempengaruhi Keselamatan Kerja

Keselamatan kerja dipengaruhi oleh beberapa faktor utama yaitu manusia, mesin, material, lingkungan, dan sistem manajemen.

1. Faktor Manusia

Faktor manusia merupakan penyebab utama kecelakaan kerja. Beberapa aspek yang mempengaruhi keselamatan kerja yaitu:

- Pengetahuan dan keterampilan pekerja
- Sikap dan perilaku terhadap keselamatan kerja
- Kondisi fisik dan mental pekerja
- Pengalaman kerja dan pelatihan

Menurut teori Domino Heinrich, sekitar 80-85% kecelakaan kerja disebabkan oleh kesalahan manusia (*human error*).

Contoh pada area tambang:

Operator alat berat yang tidak memiliki pelatihan yang memadai dapat meningkatkan risiko kecelakaan kerja.

2. Faktor Mesin atau Peralatan

Mesin yang digunakan dalam proses pertambangan harus memiliki sistem keselamatan yang baik. Faktor yang mempengaruhi keselamatan kerja dari mesin antara lain:

- Desain mesin yang aman
- Pemeliharaan mesin secara berkala
- Penggunaan alat yang sesuai dengan pekerjaan

Contoh pada denah PT Bukit Asam:

Mesin conveyor di area produksi harus dilengkapi dengan *emergency stop* untuk mencegah kecelakaan kerja.

3. Faktor Material atau Bahan

Material yang digunakan dalam proses pertambangan dapat menimbulkan bahaya jika tidak ditangani dengan benar.

Contoh material berbahaya:

- Debu batu bara
- Bahan bakar dan oli mesin
- Bahan kimia laboratorium

Penelitian menunjukkan bahwa paparan debu batu bara dalam jangka panjang dapat menyebabkan gangguan sistem pernapasan pada pekerja tambang (Aini, 2015).

4. Faktor Lingkungan Kerja

Lingkungan kerja yang tidak aman dapat meningkatkan risiko kecelakaan kerja. Beberapa faktor lingkungan yang mempengaruhi keselamatan kerja antara lain:

- Kebisingan dari alat berat
- Pencahayaan yang kurang memadai
- Suhu lingkungan yang tinggi
- Tata letak area kerja yang tidak teratur

5. Faktor Sistem Manajemen

Sistem manajemen keselamatan kerja berperan dalam mengatur seluruh aktivitas kerja agar tetap aman. Contoh implementasi:

- Kebijakan keselamatan kerja
- Program pelatihan K3
- Audit keselamatan kerja
- Partisipasi pekerja dalam program K3

2.3 Sumber Bahaya (Hazard) yang Berpotensi Menimbulkan Kecelakaan

Sumber bahaya adalah segala sesuatu yang berpotensi menyebabkan cedera, penyakit, atau kerusakan pada pekerja.

1. Bahaya Fisik (Physical Hazards)

Bahaya fisik merupakan bahaya yang berasal dari kondisi fisik lingkungan kerja maupun penggunaan peralatan kerja.

Beberapa contoh bahaya fisik antara lain:

- Kebisingan dari mesin dan alat berat
- Debu batu bara yang beterbangan saat proses penambangan
- Suhu lingkungan kerja yang tinggi
- Getaran dari peralatan tambang

Contoh dalam kegiatan pertambangan:

Aktivitas penggalian dan pengangkutan batubara menggunakan alat berat seperti excavator dan dump truck dapat menimbulkan kebisingan tinggi serta paparan debu yang berpotensi mengganggu kesehatan pekerja.

MANAGEMENT RISIKO PT BUKIT ASAM TBK					
1. BAHAYA FISIK (PHYSICAL HAZARDS)					
Contoh: Kebisingan, Debu Batu Bara, Suhu Tinggi, Getaran					
		LIKELIHOOD OF OCCURRENCE			
		Very Unlikely Little or no chance of occurrence	Unlikely Could occur, less than 50/50 chance	Possible 50/50 chance	Probable More likely to occur than not
HAZARD SEVERITY	Minor No or minor injury	CARE	CARE	CARE	CAUTION
	Moderate Off-site medical treatment or DAFW	CARE	CARE	CAUTION	ALERT
	Serious More than one DAFW – long-term absence	CARE	CAUTION	ALERT	ALARM
	Major Permanent disability or fatality	CAUTION	ALERT	ALARM	ALARM
Contoh Sumber Bahaya Fisik di PT Bukit Asam Tbk: • Kebisingan alat berat (excavator, dump truck) • Debu batu bara saat penambangan • Suhu kerja tinggi di area produksi • Getaran dari alat berat dan conveyor					

2. Bahaya Kimia (Chemical Hazards)

Bahaya kimia berasal dari paparan bahan kimia berbahaya yang digunakan atau dihasilkan selama proses kerja.

Beberapa contoh bahaya kimia antara lain:

- Debu batu bara
- Gas berbahaya yang dihasilkan dari proses tambang
- Bahan bakar dan oli mesin

- Bahan kimia yang digunakan dalam laboratorium

Contoh dalam kegiatan pertambangan:

Paparan debu batu bara dalam jangka waktu lama dapat menyebabkan gangguan pada sistem pernapasan pekerja tambang.

MANAGEMENT RISIKO PT BUKIT ASAM TBK					
2. BAHAYA KIMIA (CHEMICAL HAZARDS)					
Contoh: Debu Batubara, Gas Berbahaya, Bahan Bakar & Oli, Bahan Kimia Lab					
HAZARD SEVERITY		LIKELIHOOD OF OCCURRENCE			
		Very Unlikely Little or no chance of occurrence	Unlikely Could occur, less than 50/50 chance	Possible 50/50 chance	Probable More likely to occur than not
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	Moderate Off-site medical treatment or DAFW	CARE	CAUTION	ALERT	ALARM
	Serious More than one DAFW – long-term absence	CAUTION	ALERT	ALARM	ALARM
	Major Permanent disability or fatality	ALERT	ALARM	ALARM	ALARM
Contoh Sumber Bahaya Kimia di PT Bukit Asam Tbk: • Paparan debu batu bara • Gas berbahaya (CO, CH ₄ , H ₂ S) • Tumpahan bahan bakar dan oli mesin • Bahan kimia laboratorium					

3. Bahaya Biologi (Biological Hazards)

Bahaya biologis berasal dari organisme hidup seperti bakteri, virus, dan jamur yang dapat menyebabkan penyakit pada pekerja.

Beberapa contoh bahaya biologis antara lain:

- Penyebaran penyakit menular antar pekerja
- Lingkungan kerja yang kurang higienis
- Kontaminasi mikroorganisme pada fasilitas pekerja

Contoh dalam lingkungan pertambangan:

Pekerja yang tinggal di mess karyawan memiliki potensi tertular penyakit menular jika kebersihan lingkungan tidak dijaga dengan baik.

MANAGEMENT RISIKO PT BUKIT ASAM TBK					
3. BAHAYA BIOLOGI (BIOLOGICAL HAZARDS)					
Contoh: Penyakit Menular, Lingkungan Tidak Higienis, Kontaminasi Mikroorganisme					
HAZARD SEVERITY		LIKELIHOOD OF OCCURRENCE			
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	Serious More than one DAFW – long-term absence	CARE	CAUTION	ALERT	ALERT
	Major Permanent disability or fatality	CAUTION	ALERT	ALARM	ALARM
Contoh Sumber Bahaya Biologi di PT Bukit Asam Tbk: • Penyebaran penyakit menular di mess karyawan • Lingkungan kerja dan fasilitas kurang higienis • Kontaminasi mikroorganisme pada fasilitas umum					

4. Bahaya Ergonomi (Ergonomic Hazards)

Bahaya ergonomi berkaitan dengan desain pekerjaan, posisi kerja yang tidak sesuai, serta aktivitas kerja yang dilakukan secara berulang dalam jangka waktu lama.

Beberapa contoh bahaya ergonomi antara lain:

- Posisi duduk yang tidak ergonomis saat bekerja dengan komputer
- Aktivitas mengangkat beban berat
- Gerakan kerja yang berulang

Contoh dalam kegiatan pertambangan:

Pekerja yang melakukan pengangkatan peralatan atau material secara manual dapat mengalami kelelahan otot maupun cedera pada sistem muskuloskeletal.

MANAGEMENT RISIKO PT BUKIT ASAM TBK					
4. BAHAYA ERGONOMI (ERGONOMIC HAZARDS)					
Contoh: Angkat Manual, Posisi Kerja Tidak Ergonomis, Gerakan Berulang					
		LIKELIHOOD OF OCCURRENCE			
		Very Unlikely Little or no chance of occurrence	Unlikely Could occur, less than 50/50 chance	Possible 50/50 chance	Probable More likely to occur than not
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	Major Permanent disability or fatality	ALERT	ALERT	ALARM	ALARM
Contoh Sumber Bahaya Ergonomi di PT Bukit Asam Tbk: • Pengangkatan material secara manual • Posisi duduk operator tidak ergonomis • Gerakan kerja berulang dalam waktu lama					

5. Bahaya Psikososial (Psychosocial Hazards)

Bahaya psikososial berkaitan dengan kondisi psikologis pekerja yang dipengaruhi oleh tekanan kerja maupun lingkungan kerja.

Beberapa faktor yang dapat menimbulkan bahaya psikososial antara lain:

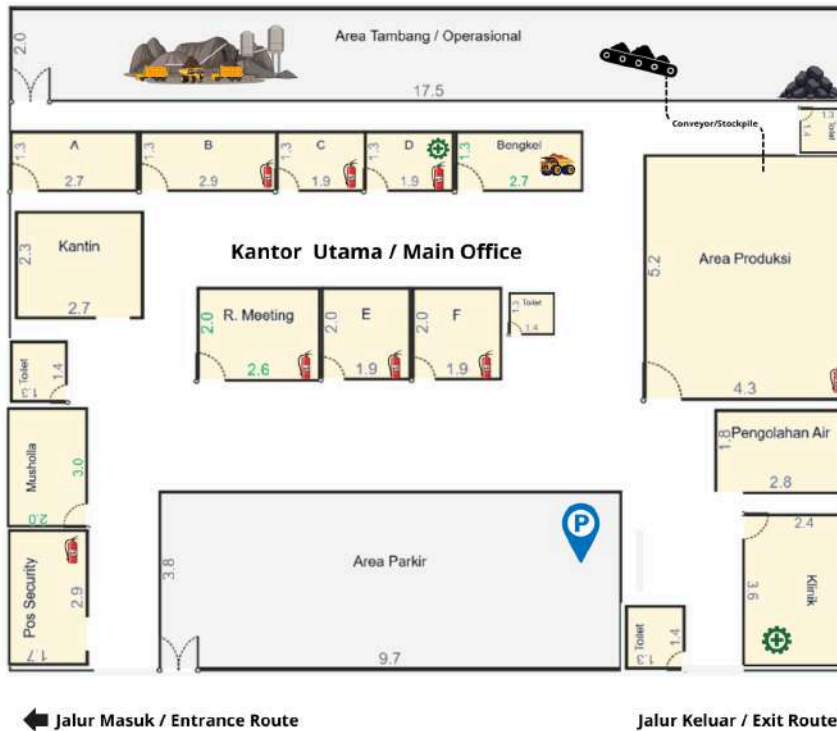
- Beban kerja yang tinggi
- Jam kerja yang panjang
- Tekanan target produksi
- Lingkungan kerja yang berisiko tinggi

Contoh dalam industri pertambangan:

Tekanan target produksi dan kondisi kerja di area tambang yang memiliki risiko tinggi dapat menimbulkan stres kerja yang berpotensi menurunkan konsentrasi pekerja.

MANAGEMENT RISIKO PT BUKIT ASAM TBK					
5. BAHAYA PSIKOSOSIAL (PSYCHOSOCIAL HAZARDS)					
Contoh: Tekanan Target Produksi, Jam Kerja Panjang, Lingkungan Kerja Berisiko					
		LIKELIHOOD OF OCCURRENCE			
		Very Unlikely Little or no chance of occurrence	Unlikely Could occur, less than 50/50 chance	Possible 50/50 chance	Probable More likely to occur than not
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	Serious More than one DAFW – long-term absence	ALERT	ALERT	ALARM	ALARM
	Major Permanent disability or fatality	ALERT	ALARM	ALARM	ALARM
Contoh Sumber Bahaya Psikososial di PT Bukit Asam Tbk: • Tekanan target produksi • Jam kerja panjang dan lembur berlebihan • Lingkungan kerja berisiko tinggi yang menimbulkan stres					

2.4 Analisis Potensi Bahaya Berdasarkan Denah PT Bukit Asam



- A: Pos Pengecekan APD
- B: Ruang Kontrol / Control Room
- C: Laboratorium Uji Kualitas Batu Bara
- D: Ruang K3 / HSE Room
- E: Ruang Teknik / Technical Room
- F: Mess Karyawan / Employee Dormitory

Area	Potensi Bahaya
Area Tambang	Kecelakaan alat berat, debu batu bara, kebisingan
Area Produksi	Mesin conveyor, suhu panas, debu
Laboratorium	Bahan kimia berbahaya
Bengkel	Percikan api, alat tajam

Kantor	Bahaya ergonomi
Mess Karyawan	Penyakit menular
Area Parkir	Kecelakaan kendaraan

Penelitian mengenai proses pengangkutan batubara menunjukkan bahwa aktivitas *coal hauling* dan *coal loading* memiliki risiko kecelakaan yang tinggi akibat penggunaan alat berat dan kendaraan tambang (Sainyakit & Djunaidi, 2023).

2.4.1 Identifikasi Sumber Bahaya pada Setiap Area Kerja

MANAGEMENT RISIKO PT BUKIT ASAM TBK 1. AREA TAMBANG / OPERASIONAL Contoh: Kebisingan, Debu Batu Bara, Suhu Tinggi, Getaran		MANAGEMENT RISIKO PT BUKIT ASAM TBK 2. AREA PRODUKSI (CRUSHING & CONVEYOR) Contoh: Debu, Kebisingan, Benda Terjepit, Terjatuh		MANAGEMENT RISIKO PT BUKIT ASAM TBK 3. BENGKEL (MAINTENANCE WORKSHOP) Contoh: Tumpahan Oli, Percikan Api, Peralatan Tajam, Listrik	
LIKELIHOOD OF OCCURRENCE Very Unlikely Little or no chance of occurrence Unlikely Could occur less than 50/50 chance Possible 50/50 chance Probable More likely to occur than not		LIKELIHOOD OF OCCURRENCE Very Unlikely Little or no chance of occurrence Unlikely Could occur less than 50/50 chance Possible 50/50 chance Probable More likely to occur than not		LIKELIHOOD OF OCCURRENCE Very Unlikely Little or no chance of occurrence Unlikely Could occur less than 50/50 chance Possible 50/50 chance Probable More likely to occur than not	
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Contoh Sumber Bahaya di Area Tambang / Operasional: • Kebisingan alat berat (excavator, dump truck) • Debu batu bara saat penambangan dan hauling • Suhu kerja tinggi di area produksi • Getaran dari alat berat dan conveyor • Risiko tertimpa material / badan 		Contoh Sumber Bahaya di Area Produksi: • Debu batu bara dari proses crushing • Kebisingan mesin crusher dan conveyor • Benda terjepit/tertarik di antara belt conveyor • Terjatuh dari ketinggian platform maintenance • Percikan/tumpahan material 		Contoh Sumber Bahaya di Bengkel: • Tumpahan oli, grease, dan bahan bakar • Percikan api saat pengelasan / cutting • Paparan bahan kimia (dissolver, solvent, pelumas) • Penggunaan alat tangan dan mesin berputar • Bahaya listrik dari peralatan 	
MANAGEMENT RISIKO PT BUKIT ASAM 4. KANTOR UTAMA / MAIN OFFICE Contoh: Ergonomi, Kebakaran, Listrik, Stres Kerja		MANAGEMENT RISIKO PT BUKIT ASAM TBK 5. MESS KARYAWAN / EMPLOYEE DORMITORY Contoh: Penyakit Menular, Kebersihan, Kebakaran, Listrik		MANAGEMENT RISIKO PT BUKIT ASAM TBK 6. AREA LAIN (PARKIR, PENGELOLAHAN AIR, KLINIK, DLL) Contoh: Kecelakaan Kendaraan, Bahan Kimia, Listrik, Medis	
LIKELIHOOD OF OCCURRENCE Very Unlikely Little or no chance of occurrence Unlikely Could occur less than 50/50 chance Possible 50/50 chance Probable More likely to occur than not		LIKELIHOOD OF OCCURRENCE Very Unlikely Little or no chance of occurrence Unlikely Could occur less than 50/50 chance Possible 50/50 chance Probable More likely to occur than not		LIKELIHOOD OF OCCURRENCE Very Unlikely Little or no chance of occurrence Unlikely Could occur less than 50/50 chance Possible 50/50 chance Probable More likely to occur than not	
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	Major Permanent disability or fatality	CAUTION	ALERT	ALARM	ALARM
Contoh Sumber Bahaya di Kantor Utama: • Posisi duduk tidak ergonomis • Risiko kebakaran (korsleting listrik) • Bahaya listrik dari peralatan kantor • Stres kerja akibat tekanan pekerjaan • Tersandung kabel / barang 		Contoh Sumber Bahaya di Mess Karyawan: • Penyakit menular (flu, ISPA, diare) • Lingkungan kurang higienis • Kebakaran akibat kompor / instalasi listrik • Tersandung / terjatuh di area kamar mandi • Masalah psikososial (konflik antar karyawan) 		Contoh Sumber Bahaya di Area Lain: • Kecelakaan kendaraan di area parkir • Paparan bahan kimia pada pengolahan air • Bahaya listrik pada instalasi • Bahaya medis di klinik (jerum santik, limbah medis) • Risiko terpelecut / jatuh 	

MANAGEMENT RISIKO PT BUKIT ASAM TBK 7. POS PENGECEKAN APD Contoh: Terpelant, Listrik, Kebakaran, Terbakar Kendaraan <table><thead><tr><th colspan="5">LIKELIHOOD OF OCCURRENCE</th></tr><tr><th></th><th>Very Unlikely Little or no chance of occurrence</th><th>Unlikely Small chance of occurrence</th><th>Possible Moderate chance of occurrence</th><th>Probable More likely to occur than not</th></tr></thead><tbody><tr><td>Minor Non-injury</td><td>CARE</td><td>CARE</td><td>CARE</td><td>CAUTION</td></tr><tr><td>Moderate Off-site material released to public</td><td>CARE</td><td>CARE</td><td>CAUTION</td><td>ALERT</td></tr><tr><td>Service Non-hazardous liquid spillage</td><td>CARE</td><td>CAUTION</td><td>ALERT</td><td>ALARM</td></tr><tr><td>Major Personnel injury or fatality</td><td>CAUTION</td><td>ALERT</td><td>ALARM</td><td>ALARM</td></tr></tbody></table> Contoh Sumber Bahaya di Pos Pengawasan APD: - Terpelant area basah - Koneksi listrik - Kebakaran - Terbakar kendaraan operasional - APD tidak sesuai / cacat	LIKELIHOOD OF OCCURRENCE						Very Unlikely Little or no chance of occurrence	Unlikely Small chance of occurrence	Possible Moderate chance of occurrence	Probable More likely to occur than not	Minor Non-injury	CARE	CARE	CARE	CAUTION	Moderate Off-site material released to public	CARE	CARE	CAUTION	ALERT	Service Non-hazardous liquid spillage	CARE	CAUTION	ALERT	ALARM	Major Personnel injury or fatality	CAUTION	ALERT	ALARM	ALARM	MANAGEMENT RISIKO PT BUKIT ASAM TBK 8. RUANG KONTROL / CONTROL ROOM Contoh: Listrik, Kebakaran, Kebocoran, Gangguan Sistem <table><thead><tr><th colspan="5">LIKELIHOOD OF OCCURRENCE</th></tr><tr><th></th><th>Very Unlikely Little or no chance of occurrence</th><th>Unlikely Small chance of occurrence</th><th>Possible Moderate chance of occurrence</th><th>Probable More likely to occur than not</th></tr></thead><tbody><tr><td>Minor Non-injury</td><td>CARE</td><td>CARE</td><td>CARE</td><td>CAUTION</td></tr><tr><td>Moderate Off-site material released to public</td><td>CARE</td><td>CARE</td><td>CAUTION</td><td>ALERT</td></tr><tr><td>Service Non-hazardous liquid spillage</td><td>CARE</td><td>CAUTION</td><td>ALERT</td><td>ALARM</td></tr><tr><td>Major Personnel injury or fatality</td><td>CAUTION</td><td>ALERT</td><td>ALARM</td><td>ALARM</td></tr></tbody></table> Contoh Sumber Bahaya di Ruang Kontrol: - Gangguan listrik / pemadaman - Kebakaran akibat peralatan listrik - Kebocoran / tumpukan - Gangguan ekspansi / software - Tumpukan instrumen / dipi	LIKELIHOOD OF OCCURRENCE						Very Unlikely Little or no chance of occurrence	Unlikely Small chance of occurrence	Possible Moderate chance of occurrence	Probable More likely to occur than not	Minor Non-injury	CARE	CARE	CARE	CAUTION	Moderate Off-site material released to public	CARE	CARE	CAUTION	ALERT	Service Non-hazardous liquid spillage	CARE	CAUTION	ALERT	ALARM	Major Personnel injury or fatality	CAUTION	ALERT	ALARM	ALARM	MANAGEMENT RISIKO PT BUKIT ASAM TBK 9. LABORATORIUM UJI KUALITAS BATU BARA Contoh: Bahan Kimia, Api, Pencucian Kaca, Paparan Debu <table><thead><tr><th colspan="5">LIKELIHOOD OF OCCURRENCE</th></tr><tr><th></th><th>Very Unlikely Little or no chance of occurrence</th><th>Unlikely Small chance of occurrence</th><th>Possible Moderate chance of occurrence</th><th>Probable More likely to occur than not</th></tr></thead><tbody><tr><td>Minor Non-injury</td><td>CARE</td><td>CARE</td><td>CAUTION</td><td>CAUTION</td></tr><tr><td>Moderate Off-site material released to public</td><td>CARE</td><td>CAUTION</td><td>ALERT</td><td>ALERT</td></tr><tr><td>Service Non-hazardous liquid spillage</td><td>CAUTION</td><td>ALERT</td><td>ALARM</td><td>ALARM</td></tr><tr><td>Major Personnel injury or fatality</td><td>ALERT</td><td>ALARM</td><td>ALARM</td><td>ALARM</td></tr></tbody></table> Contoh Sumber Bahaya di Laboratorium: - Paparan bahan kimia berbahaya - Pecahan kaca alat laboratorium - Kebakaran akibat tumpukan mudah terbakar - Paparan debu batu bara - Tumpahannya bahan kimia	LIKELIHOOD OF OCCURRENCE						Very Unlikely Little or no chance of occurrence	Unlikely Small chance of occurrence	Possible Moderate chance of occurrence	Probable More likely to occur than not	Minor Non-injury	CARE	CARE	CAUTION	CAUTION	Moderate Off-site material released to public	CARE	CAUTION	ALERT	ALERT	Service Non-hazardous liquid spillage	CAUTION	ALERT	ALARM	ALARM	Major Personnel injury or fatality	ALERT	ALARM	ALARM	ALARM	MANAGEMENT RISIKO PT BUKIT ASAM TBK 10. RUANG K3 / HSE ROOM Contoh: Listrik, Kebakaran, Arasp Jatuh, Ergonomi <table><thead><tr><th colspan="5">LIKELIHOOD OF OCCURRENCE</th></tr><tr><th></th><th>Very Unlikely Little or no chance of occurrence</th><th>Unlikely Small chance of occurrence</th><th>Possible Moderate chance of occurrence</th><th>Probable More likely to occur than not</th></tr></thead><tbody><tr><td>Minor Non-injury</td><td>CARE</td><td>CARE</td><td>CARE</td><td>CAUTION</td></tr><tr><td>Moderate Off-site material released to public</td><td>CARE</td><td>CARE</td><td>CAUTION</td><td>ALERT</td></tr><tr><td>Service Non-hazardous liquid spillage</td><td>CARE</td><td>CAUTION</td><td>ALERT</td><td>ALARM</td></tr><tr><td>Major Personnel injury or fatality</td><td>CAUTION</td><td>ALERT</td><td>ALARM</td><td>ALARM</td></tr></tbody></table> Contoh Sumber Bahaya di Ruang K3 / HSE: - Kebakaran akibat koneksi listrik - Arasp / Dokumen jatuh - Ergonomi tidak ergonomis - Tersandung kabel / barang - Ketersinggungan ventilasi	LIKELIHOOD OF OCCURRENCE						Very Unlikely Little or no chance of occurrence	Unlikely Small chance of occurrence	Possible Moderate chance of occurrence	Probable More likely to occur than not	Minor Non-injury	CARE	CARE	CARE	CAUTION	Moderate Off-site material released to public	CARE	CARE	CAUTION	ALERT	Service Non-hazardous liquid spillage	CARE	CAUTION	ALERT	ALARM	Major Personnel injury or fatality	CAUTION	ALERT	ALARM	ALARM	MANAGEMENT RISIKO PT BUKIT ASAM TBK 11. RUANG TEKNIK / TECHNICAL ROOM Contoh: Listrik, Perawatan Tajam, Terjatuh, Kebakaran <table><thead><tr><th colspan="5">LIKELIHOOD OF OCCURRENCE</th></tr><tr><th></th><th>Very Unlikely Little or no chance of occurrence</th><th>Unlikely Small chance of occurrence</th><th>Possible Moderate chance of occurrence</th><th>Probable More likely to occur than not</th></tr></thead><tbody><tr><td>Minor Non-injury</td><td>CARE</td><td>CARE</td><td>CAUTION</td><td>CAUTION</td></tr><tr><td>Moderate Off-site material released to public</td><td>CARE</td><td>CAUTION</td><td>ALERT</td><td>ALERT</td></tr><tr><td>Service Non-hazardous liquid spillage</td><td>CARE</td><td>CAUTION</td><td>ALERT</td><td>ALARM</td></tr><tr><td>Major Personnel injury or fatality</td><td>ALERT</td><td>ALARM</td><td>ALARM</td><td>ALARM</td></tr></tbody></table> Contoh Sumber Bahaya di Ruang Teknik: - Koneksi listrik - Penggunaan peralatan tajam - Terjatuh mesin / komponen - Kebakaran akibat oli / bahan mudah terbakar - Tersandung alat / kabel	LIKELIHOOD OF OCCURRENCE						Very Unlikely Little or no chance of occurrence	Unlikely Small chance of occurrence	Possible Moderate chance of occurrence	Probable More likely to occur than not	Minor Non-injury	CARE	CARE	CAUTION	CAUTION	Moderate Off-site material released to public	CARE	CAUTION	ALERT	ALERT	Service Non-hazardous liquid spillage	CARE	CAUTION	ALERT	ALARM	Major Personnel injury or fatality	ALERT	ALARM	ALARM	ALARM
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MANAGEMENT RISIKO PT BUKIT ASAM TBK 12. KANTIN Contoh: Kebakaran, Kebakaran, Gas LPG, Tersedak <table><thead><tr><th colspan="5">LIKELIHOOD OF OCCURRENCE</th></tr><tr><th></th><th>Very Unlikely Little or no chance of occurrence</th><th>Unlikely Small chance of occurrence</th><th>Possible Moderate chance of occurrence</th><th>Probable More likely to occur than not</th></tr></thead><tbody><tr><td>Minor Non-injury</td><td>CARE</td><td>CARE</td><td>CARE</td><td>CAUTION</td></tr><tr><td>Moderate Off-site material released to public</td><td>CARE</td><td>CARE</td><td>CAUTION</td><td>ALERT</td></tr><tr><td>Service Non-hazardous liquid spillage</td><td>CARE</td><td>CAUTION</td><td>ALERT</td><td>ALARM</td></tr><tr><td>Major Personnel injury or fatality</td><td>CAUTION</td><td>ALERT</td><td>ALARM</td><td>ALARM</td></tr></tbody></table> Contoh Sumber Bahaya di Kantin: - Kebakaran makanan - Kebakaran akibat kompor / gas LPG - Tersedak makanan / minuman - Lantai licin / tersandung - Kebakaran peralatan makan	LIKELIHOOD OF OCCURRENCE						Very Unlikely Little or no chance of occurrence	Unlikely Small chance of occurrence	Possible Moderate chance of occurrence	Probable More likely to occur than not	Minor Non-injury	CARE	CARE	CARE	CAUTION	Moderate Off-site material released to public	CARE	CARE	CAUTION	ALERT	Service Non-hazardous liquid spillage	CARE	CAUTION	ALERT	ALARM	Major Personnel injury or fatality	CAUTION	ALERT	ALARM	ALARM	MANAGEMENT RISIKO PT BUKIT ASAM TBK 13. PENGOLAHAN AIR Contoh: Bahan Kimia, Listrik, Terpelant, Paparan Mikroba <table><thead><tr><th colspan="5">LIKELIHOOD OF OCCURRENCE</th></tr><tr><th></th><th>Very Unlikely Little or no chance of occurrence</th><th>Unlikely Small chance of occurrence</th><th>Possible Moderate chance of occurrence</th><th>Probable More likely to occur than not</th></tr></thead><tbody><tr><td>Minor Non-injury</td><td>CARE</td><td>CARE</td><td>CARE</td><td>CAUTION</td></tr><tr><td>Moderate Off-site material released to public</td><td>CARE</td><td>CAUTION</td><td>ALERT</td><td>ALERT</td></tr><tr><td>Service Non-hazardous liquid spillage</td><td>CAUTION</td><td>ALERT</td><td>ALARM</td><td>ALARM</td></tr><tr><td>Major Personnel injury or fatality</td><td>ALERT</td><td>ALARM</td><td>ALARM</td><td>ALARM</td></tr></tbody></table> Contoh Sumber Bahaya di Pengolahan Air: - Paparan bahan kimia (klorin, kalsium) - Koneksi listrik pada pompa / panel - Terpelant area basah / licin - Paparan mikroba / kontaminasi air - Kebocoran pipa / tumpukan bahan kimia	LIKELIHOOD OF OCCURRENCE						Very Unlikely Little or no chance of occurrence	Unlikely Small chance of occurrence	Possible Moderate chance of occurrence	Probable More likely to occur than not	Minor Non-injury	CARE	CARE	CARE	CAUTION	Moderate Off-site material released to public	CARE	CAUTION	ALERT	ALERT	Service Non-hazardous liquid spillage	CAUTION	ALERT	ALARM	ALARM	Major Personnel injury or fatality	ALERT	ALARM	ALARM	ALARM	MANAGEMENT RISIKO PT BUKIT ASAM TBK 14. KLINIK Contoh: Medis, Infeksi, Listrik, Kebakaran <table><thead><tr><th colspan="5">LIKELIHOOD OF OCCURRENCE</th></tr><tr><th></th><th>Very Unlikely Little or no chance of occurrence</th><th>Unlikely Small chance of occurrence</th><th>Possible Moderate chance of occurrence</th><th>Probable More likely to occur than not</th></tr></thead><tbody><tr><td>Minor Non-injury</td><td>CARE</td><td>CARE</td><td>CARE</td><td>CAUTION</td></tr><tr><td>Moderate Off-site material released to public</td><td>CARE</td><td>CARE</td><td>CAUTION</td><td>ALERT</td></tr><tr><td>Service Non-hazardous liquid spillage</td><td>CARE</td><td>CAUTION</td><td>ALERT</td><td>ALARM</td></tr><tr><td>Major Personnel injury or fatality</td><td>CAUTION</td><td>ALERT</td><td>ALARM</td><td>ALARM</td></tr></tbody></table> Contoh Sumber Bahaya di Klinik: - Infeksi / penyakit menular - Penggunaan alat medis tajam - Koneksi listrik peralatan medis - Kebakaran - Penggunaan obat / bahan kimia medis	LIKELIHOOD OF OCCURRENCE						Very Unlikely Little or no chance of occurrence	Unlikely Small chance of occurrence	Possible Moderate chance of occurrence	Probable More likely to occur than not	Minor Non-injury	CARE	CARE	CARE	CAUTION	Moderate Off-site material released to public	CARE	CARE	CAUTION	ALERT	Service Non-hazardous liquid spillage	CARE	CAUTION	ALERT	ALARM	Major Personnel injury or fatality	CAUTION	ALERT	ALARM	ALARM	MANAGEMENT RISIKO PT BUKIT ASAM TBK 15. AREA PARKIR Contoh: Tabrakan, Terpelant, Pencurian, Panas Matahari <table><thead><tr><th colspan="5">LIKELIHOOD OF OCCURRENCE</th></tr><tr><th></th><th>Very Unlikely Little or no chance of occurrence</th><th>Unlikely Small chance of occurrence</th><th>Possible Moderate chance of occurrence</th><th>Probable More likely to occur than not</th></tr></thead><tbody><tr><td>Minor Non-injury</td><td>CARE</td><td>CARE</td><td>CARE</td><td>CAUTION</td></tr><tr><td>Moderate Off-site material released to public</td><td>CARE</td><td>CARE</td><td>CAUTION</td><td>ALERT</td></tr><tr><td>Service Non-hazardous liquid spillage</td><td>CARE</td><td>CAUTION</td><td>ALERT</td><td>ALARM</td></tr><tr><td>Major Personnel injury or fatality</td><td>CAUTION</td><td>ALERT</td><td>ALARM</td><td>ALARM</td></tr></tbody></table> Contoh Sumber Bahaya di Area Parkir: - Tabrakan kendaraan - Terpelant / jatuh - Pencurian kendaraan / barang - Panas matahari / dehidrasi - Pemencaran kurang	LIKELIHOOD OF OCCURRENCE						Very Unlikely Little or no chance of occurrence	Unlikely Small chance of occurrence	Possible Moderate chance of occurrence	Probable More likely to occur than not	Minor Non-injury	CARE	CARE	CARE	CAUTION	Moderate Off-site material released to public	CARE	CARE	CAUTION	ALERT	Service Non-hazardous liquid spillage	CARE	CAUTION	ALERT	ALARM	Major Personnel injury or fatality	CAUTION	ALERT	ALARM	ALARM	MANAGEMENT RISIKO PT BUKIT ASAM TBK 16. POS SECURITY Contoh: Listrik, Kebakaran, Terawang Hewan, Fatigue <table><thead><tr><th colspan="5">LIKELIHOOD OF OCCURRENCE</th></tr><tr><th></th><th>Very Unlikely Little or no chance of occurrence</th><th>Unlikely Small chance of occurrence</th><th>Possible Moderate chance of occurrence</th><th>Probable More likely to occur than not</th></tr></thead><tbody><tr><td>Minor Non-injury</td><td>CARE</td><td>CARE</td><td>CARE</td><td>CAUTION</td></tr><tr><td>Moderate Off-site material released to public</td><td>CARE</td><td>CARE</td><td>CAUTION</td><td>ALERT</td></tr><tr><td>Service Non-hazardous liquid spillage</td><td>CARE</td><td>CAUTION</td><td>ALERT</td><td>ALARM</td></tr><tr><td>Major Personnel injury or fatality</td><td>CAUTION</td><td>ALERT</td><td>ALARM</td><td>ALARM</td></tr></tbody></table> Contoh Sumber Bahaya di Pos Security: - Koneksi listrik - Kebakaran - Terawang hewan / binatang - Infeksi / kurangnya istirahat - Kebocoran / tumpukan	LIKELIHOOD OF OCCURRENCE						Very Unlikely Little or no chance of occurrence	Unlikely Small chance of occurrence	Possible Moderate chance of occurrence	Probable More likely to occur than not	Minor Non-injury	CARE	CARE	CARE	CAUTION	Moderate Off-site material released to public	CARE	CARE	CAUTION	ALERT	Service Non-hazardous liquid spillage	CARE	CAUTION	ALERT	ALARM	Major Personnel injury or fatality	CAUTION	ALERT	ALARM	ALARM
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KETERANGAN LEVEL RISIKO CARE Risiko rendah dapat ditangani dengan penanganan rutin CAUTION Risiko menengah perlu pengendalian untuk mencegah insiden ALERT Risiko tinggi perlu tindakan kontrol segera ALARM Risiko sangat tinggi perlu tindakan segera sebelum penanganan dilakukan Bayangan Listrik Tabrakan Kendaraan Bahaya Kimia Bahaya Jatuh Bahaya Gas Bahaya Keamanan Bahaya Hewan Bahaya Lelah																																																																																																																																																										

BAB III

PENUTUP

Kesimpulan

Berdasarkan analisis denah PT Bukit Asam Tbk sektor pertambangan batu bara, dapat disimpulkan bahwa:

1. Pencegahan kecelakaan kerja dapat dilakukan melalui penerapan sistem manajemen K3, pelatihan pekerja, serta pengendalian bahaya.
2. Faktor yang mempengaruhi keselamatan kerja meliputi faktor manusia, mesin, material, lingkungan kerja, dan sistem manajemen.
3. Sumber bahaya di lingkungan pertambangan meliputi bahaya fisik, kimia, biologi, ergonomi, dan psikososial.
4. Identifikasi bahaya berdasarkan denah perusahaan sangat penting untuk mencegah kecelakaan kerja di lingkungan pertambangan.

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